

Geography and convergence

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Abstract

Economic convergence across European countries is a key objective of European policy, but has stalled since around 1990. We argue, using insights from economic geography, that rather than being surprised by this, we should actually expect convergence to be limited. We show that despite many political and historical differences, income dispersion across US states and European countries has been remarkably similar over the last century. A theory of income differences across space due to agglomeration, geographic differences and migration, provides a way to understand this. (JEL: E23,F15,F22)

1. Introduction

For eighty years since the end of World War II, European policy has pursued integration with a view to achieving greater economic equality across the continent. The core treaty of the European Union states that: “the Union shall aim at reducing disparities between the levels of development of the various regions and the backwardness of the least favoured regions.”¹ In terms of policy, there have been many achievements. Free trade, free movement of people, a common currency, and centralized monetary policy. However, outcomes have been somewhat disappointing, as there are still large disparities in income per capita across countries. For example, in 2018, real income per capita in Greece and Portugal was only 49% and 57% as high as in the Netherlands, respectively.

In this paper, we revisit the study of convergence from the perspective of economic geography. In recent decades, this field has emerged as a vibrant area of economics research, studying the forces shaping the distribution of economic activity across different regions and cities within a country. Strikingly, this literature sees inequalities

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1. Article 174 of the Consolidated Treaty of the European Union.

across a country as a natural equilibrium outcome and provides a theory to explain why. If inequality across regions within a country is natural, then why not across countries within a continent? From the perspective of this field, we should not be surprised that some countries in Europe are poorer than others. We should *expect* it.

We make this point empirically and theoretically. Empirically, we compare the experiences of the United States (US) and Europe over the last century. We think of the US as an example of an economic union with a roughly similar level of economic development to Europe, but with a higher level of economic integration.² Our prior is therefore that, to the extent that further economic integration in Europe could generate greater convergence in income per capita, we should see less dispersion in the US than in Europe.

This is not what we find. Since around 1990, income dispersion across US states and European countries has been quite similar. Moreover, it has followed similar trajectories over the course of the 20th century in these areas. There were similar levels of dispersion before World War II (WWII), and convergence since then, with the main difference being that WWII appears to have delayed the start of convergence in Europe by about a decade.

If the US is highly economically integrated yet still has substantial income dispersion across states, what are the economic reasons for this? And what are the implications for Europe? We consider these questions using ideas from economic geography. We take a standard model of production at the regional level (a state or a country) and add three twists. Regions have immutable characteristics that affect how much they can produce, for example, their amount of usable land or other geographic characteristics such as access to rivers or oceans. There is also an agglomeration force that generates productivity gains from having more people located in one region, and people can migrate, although this is costly.³ The rationale for agglomeration is that it is easier to exchange ideas and knowledge when people are closer together.

We take a simple classical view of convergence as a benchmark. Under this, technology will be shared across countries over time, and poorer countries will invest to grow their capital stock. If technology is fully shared, capital investment eventually leads to equal per capita incomes across countries. Once there are different fixed factors of production across countries, or unequal population and agglomeration, this is no longer true. These factors result in productivity differences across countries, even if they have access to the same technology, thereby generating income per capita differences. In other words, considering geography takes us away from a world in which convergence is the natural outcome.

2. For example, as well as having free trade, free movement of people, a common currency and centralized monetary policy, the US also has a more centralized political and fiscal system, a common language, and less cultural heterogeneity.

3. We study an economy for which there are positive gains from agglomeration. The gains decrease with population, but we abstract from the question of the limit of these gains and when they turn negative. The implicit assumption is that the economy starts in a region in which the gains are positive, and that the amount of migration does not exhaust the gains.

Migration offsets these forces towards divergence. When people can move between regions, they migrate towards a higher income region. Due to the fixed factor of production, there are decreasing returns to scale, and the wage falls as more people enter a region. If migration is costless, then convergence is restored, as people move to a higher income region until the income difference is eliminated. If it is costly—the empirically relevant case—then permanent income differences are sustainable, with the income gap across regions depending on the magnitude of the cost.

Agglomeration has two relevant effects. First, if the gains from concentrating people in a region are stronger, then in equilibrium there will be more migration. The observed income difference will not change though, as this is determined by the migration cost. Second, agglomeration creates an externality because firms do not internalize that when they demand more labor, it brings more people to their region and increases the productivity of everyone else. Thus, a planner who internalizes the agglomeration force would want more migration than in the competitive equilibrium, and smaller income gaps across regions.

From the perspective of this theory, the post-war period of convergence observed in the US and Europe can be interpreted as a result of technology disseminating across space, capital accumulation in poorer regions, and migration to higher income regions. But this process has limits due to geographic differences between regions that affect their output, agglomeration encouraging population concentration, and migration costs. The data looks like these limits were reached at around 1980 in the US, and 1990 in Europe.

In the final section of the paper, we provide empirical evidence consistent with migration towards wealthier regions within Europe, document greater population concentration in the US than Europe, and interpret this with the model.

Related literature. We study economic convergence using ideas from the New Economic Geography literature. Seminal papers in this field include Fujita (1988); Krugman (1991); Venables (1996), and there have been numerous contributions since. In particular, we use the ideas that agglomeration forces generate economic gains from a more concentrated population, and that geographic differences between locations affect production. Redding and Venables (2004) provide evidence that geographic differences matter for economic outcomes. Research also supports the relevance of agglomeration. Henderson (2003) argues that urban concentration has a quantitatively important effect on productivity growth, and Ciccone (2002) finds that agglomeration has positive effects in Europe, with a similar magnitude to in the US. Migration goes hand in hand with agglomeration, and some papers have examined its role in convergence. Ganong and Shoag (2017) study the connection between population growth and convergence in the US since WWII. Huber and Tondl (2012) argue that migration did not generate convergence in Europe between 2000 and 2007. Our data indicate that convergence in Europe had stalled by this time.

There is existing work considering regional economic differences in Europe using the perspective of New Economic Geography. In early contributions Puga (1999, 2002) consider regional inequality and regional policy expenditures. In a more recent example, Caliendo *et al.* (2021) use a quantitative model with agglomeration forces and migration to study the effects of changes to European integration policy in 2004. We provide a

simple stylized model, which focuses on cross-regional income convergence and nests neoclassical theory.

In their seminal work on regional convergence, Barro and Sala-i Martin (1992) showed that income dispersion across US states decreased for approximately a century up to the late 1970s. Building on this, there is an extensive literature studying convergence in both the US and Europe.⁴ Our empirical analysis contributes to this literature by directly comparing income convergence in the US and Europe over the last century. The closest analysis is in Franks *et al.* (2018) and Alcidi (2019), although their time series for Europe are shorter and they only compare the US and Europe in recent years.

2. Convergence in Europe and the US

In this section, we compare the trajectories of convergence across European countries and across US states in the 20th century.

2.1. Data Description

The analysis in this section focuses on the convergence of income or output per capita across regions over time.⁵ We use “regions” as a general term to describe geographic areas within the US or Europe, with the level of disaggregation depending on the data being used. For the United States, we focus on the state level and use real personal income per capita. For Europe, we focus on countries and use real GDP per capita.⁶

For the European countries, we use the Maddison Project Database 2020 that provides measures of real GDP per capita in purchasing power parity terms for 1920–2018.⁷ We omit the period around the Second World War (WWII), from 1939 to 1949, because there were large changes in income per capita that were unrelated to the economic factors we study. We have chosen 1950 as the first post-war year to include because, by this time, GDP per capita had returned to close to its pre-war level in most countries in our sample.

4. For the US see Carlino and Mills (1993), Mitchener and McLean (1999) and Ganong and Shoag (2017). For Europe see Cuadrado-Roura (2001), Yin *et al.* (2003), Ramajo *et al.* (2008), Borsi and Metiu (2015) and Rosés and Wolf (2021). Sala-i Martin (1996) presents evidence on convergence in both the US and Europe. We abstract from the roles of capital accumulation and investment financed by European structural funds in convergence, which are the focus of Kutun and Yigit (2007); Ramajo *et al.* (2008); Becker *et al.* (2012).

5. For present purposes, we are not focused on the distinction between these. They are closely related and the choice between them is driven by data availability.

6. While the analysis is for real income per capita and real GDP per capita, we will sometimes omit the per capita qualifier and refer to them both as measures of income. It would be ideal to have equivalent geographic units for the two areas, with similar characteristics at the start of the analysis. Due to data limitations and different initial conditions, this is not possible and is a limitation of the study. To partially address this, in the Appendix we show that our results are robust to using Census divisions which have an average population that is similar to the European countries in our sample.

7. A detailed description of the data is provided by Bolt and van Zanden (2020).

Our sample includes most Western European countries: Austria, Belgium, Cyprus,⁸ Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Portugal, Spain, Sweden, and the United Kingdom. More recent members of the European Union are excluded since many do not have reliable GDP data before the 1990s, and we are interested in long-run convergence among European countries that have been jointly pursuing integration policies during the 20th century. We also exclude Luxembourg, Ireland, Norway, and Switzerland because their income per capita levels are outliers during periods of the sample, due to their country-specific circumstances.⁹

Personal income per capita for US states is measured using two datasets from the Bureau of Economic Analysis (BEA) covering 1929–2024: annual nominal personal income and population for each state. To convert nominal income into real terms, we use the implicit price deflator for the entire US GDP, also from the BEA.¹⁰ Dividing income by population gives per capita measures.¹¹

We omit Alaska and the District of Columbia (DC) from the analysis due to the idiosyncrasies of their economies and their small population sizes. Since a large share of the DC economy is the federal government, while Alaska is unique because of its dependence on income from natural resources, income in these regions is not necessarily driven by the economic forces that we focus on.

2.2. *Convergence in Europe and the US*

To evaluate convergence in Europe over time, we compute the level of GDP per capita for each country relative to a reference country, the Netherlands, with the results in Figure 1.¹² The black dashed line represents the normalized income level of the Netherlands, and all other values are relative to this.

There are three patterns that we want to highlight. First, there has been convergence of relative income over time. For example, in 1920, the lowest income countries—Portugal, Greece, Finland, and Spain—had income per capita of 30%–50% of the level of the Netherlands. By 2018, this had increased to 50–70%, with Finland now substituted by Cyprus. Second, convergence appears to have been continuous up until around 2000, with the lines trending closer and closer together.¹³ However, after this convergence appears to have stalled, and even increased during the European sovereign debt crisis

8. The data for Cyprus only begins in 1950.

9. For example, Ireland has pursued a very different corporate tax policy that has resulted in significant differences between output and income, while Norway has much higher revenue from natural resources compared to other European countries.

10. It would be preferable to use state-level price deflators; however, these are not available.

11. To the extent that prices differ across states, but these differences are stable over time, we will mismeasure the level of income dispersion but not the trend in this dispersion. A key assumption for the US analysis is therefore that relative prices across states have not changed substantially over time.

12. The Netherlands is a good reference country because it had a relatively high GDP per capita throughout the sample period and did not have sudden large income changes with respect to other countries.

13. From 1990 to 2000, the convergence looks slightly different, with the Netherlands becoming more of an outlier and other countries moving closer together.

in the early 2010s. Also, substantial differences in income per capita remain. As of 2018, income per capita in Greece, Cyprus, and Portugal is only 49%, 57%, and 57% as large as in the Netherlands, respectively. The third striking feature of the figure is that Cyprus, Greece, Portugal and Spain have had substantially lower income per capita than other countries since 1950. Although this gap has closed, it remains large even in 2018. This income gap was present for Portugal before WWII, while Spain and Greece suffered large income per capita losses during 1936-1949 and have since shared Portugal's trajectory.

For the US, we analyze the data in the same way in Figure 2, using New York as the reference state. Like the Netherlands, New York has been a relatively high-income state throughout the sample period. Since there are too many states for individual labels, we present each state with a gray line, and focus on the blue standard deviation bands to summarize how income dispersion has evolved. The solid blue line is the mean level of income relative to New York each year, and the darker and lighter blue bands are for one and two standard deviations of the cross-sectional distribution from the mean each year.¹⁴ There are two key features of the convergence of US states. First, we see that there has been gradual convergence over time, as mean relative income increased from 55% in 1929 to 83% in 2024. Second, all of the convergence appears to have happened by 1980. After this, the standard deviation bands do not narrow, and the mean does not converge towards one.

Figures 1 and 2 are strikingly similar. They both show a pattern of gradual income convergence over many decades, which eventually stops. The main difference is that this occurred in the US 20–25 years earlier than in Europe.¹⁵ To formalize this observation, we compute the coefficient of variation (CV) for cross-regional income per capita year by year for the two areas. These CVs are plotted in Figure 3, with the US and Europe in blue and orange, respectively.¹⁶ The figure confirms the impression that both the US and Europe experienced convergence over time, which ended by 1980 in the US and at around 2000 in Europe. However, the figure provides an even stronger result, that quantitatively the level of income dispersion has followed a similar trajectory in the two areas. Before WWII, the CVs were both around 0.35 and have been similar from 2000 onwards, with the caveat that income has diverged across Europe since around 2010.¹⁷ The main difference between the two areas is that there was a substantial fall in income dispersion in the US from 1939–50, while for Europe it increased a little. This gap gradually closed during the second half of the 20th century.¹⁸

14. All states are weighted equally for these calculations.

15. In the Appendix, we perform a robustness exercise to address concerns about US states and European countries having different populations on average by using US Census divisions instead of US states.

16. In the Appendix, we show that the results are robust to weighting each region by its population share within the US or Europe, and that the results are not driven by the smaller sample size for European countries.

17. There is no statistically significant difference between the CVs prior to WWII or after 1980.

18. There are, of course, differences between the economic experiences of the US and Europe over the last century. For example, Europe was more directly affected by WWII, the subsequent division between

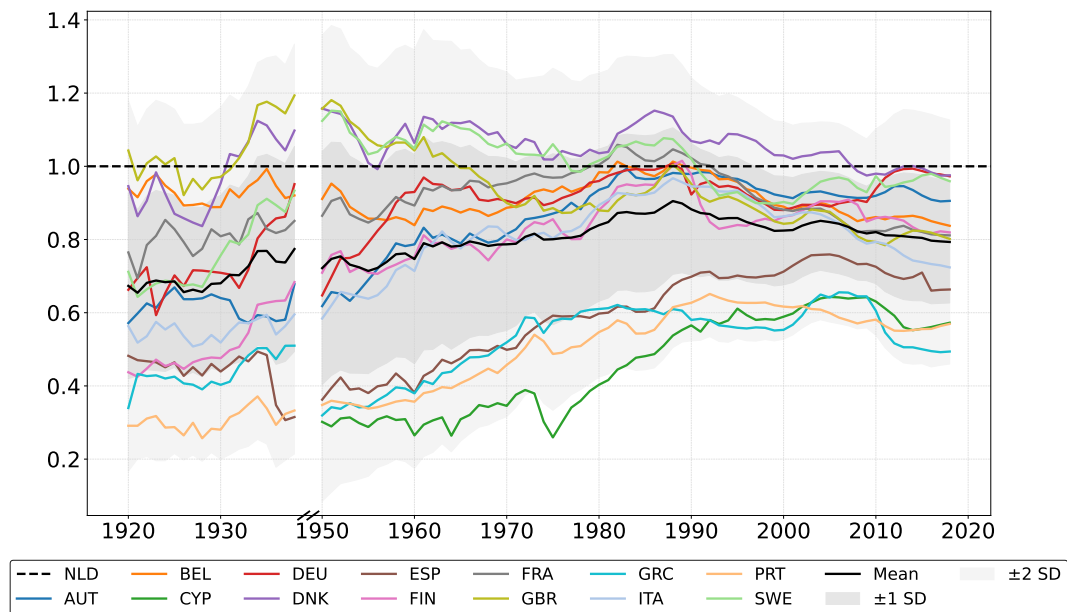


FIGURE 1: European country GDP per capita relative to the Netherlands.

Notes: GDP per capita for each country is real, PPP, and relative to the Netherlands in the relevant year. The countries are Austria (AUT), Belgium (BEL), Cyprus (CYP), Germany (DEU), Denmark (DNK), Spain (ESP), Finland (FIN), France (FRA), United Kingdom (GBR), Greece (GRC), Italy (ITA), Portugal (PRT), Sweden (SWE) and the Netherlands (NLD). The data for 1939 to 1949 are omitted. The black line is the cross-country mean, and the shaded areas are ± 1 and ± 2 standard deviations, respectively.

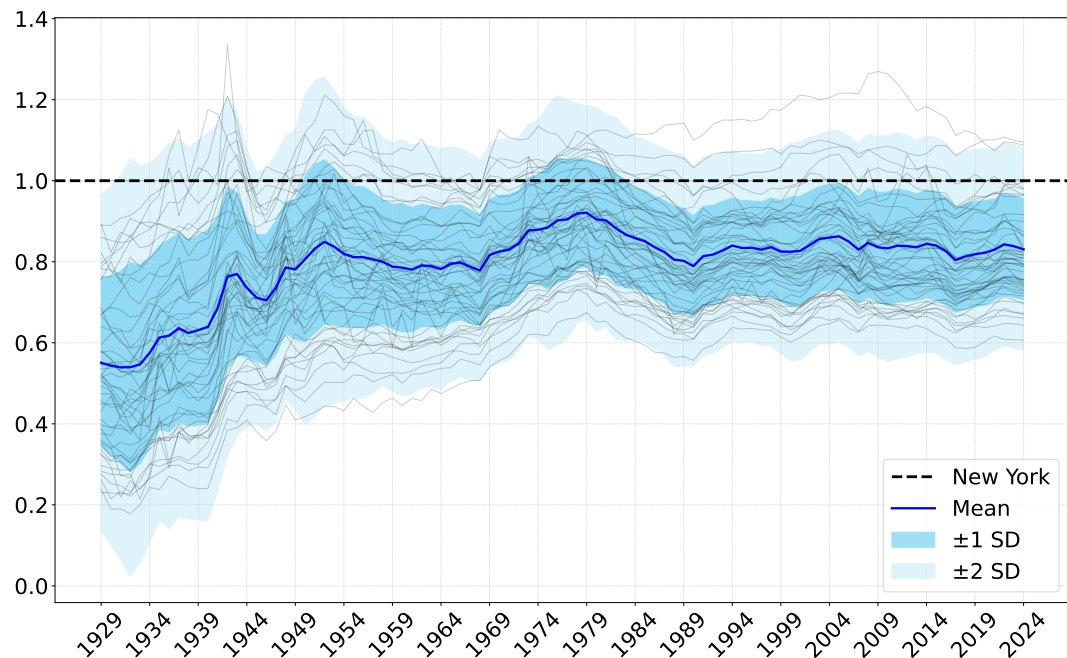


FIGURE 2: US State personal income per capita relative to New York.

Notes: Real person income per capita for each state is plotted with a light gray line. All values are relative to New York for the relevant year. The solid blue line is the cross-state mean each year, and the shaded regions are ± 1 and ± 2 standard deviations, respectively. The District of Columbia and Alaska are excluded.

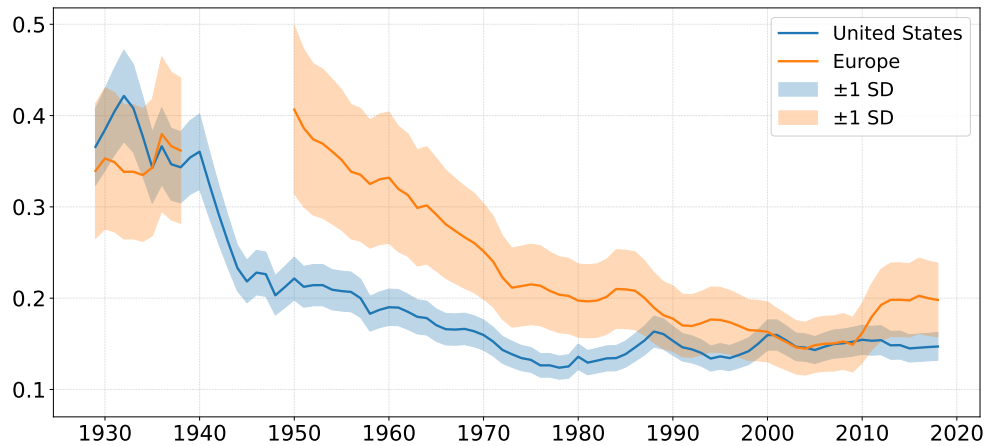


FIGURE 3: Coefficient of variation for US state personal income per capita and European country GDP per capita.

Notes: Personal income is real and GDP is real PPP. The coefficients of variation are computed year by year. For the US, the District of Columbia and Alaska are excluded. The European countries are Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Portugal, Spain, Sweden and the United Kingdom. The shaded areas are plus and minus one standard deviation confidence bands, computed following Feltz and Miller (1996) as $CV \sqrt{(1/2 + CV^2)/n}$, where CV is the coefficient of variation.

These empirical results raise several questions for European policy efforts to foster economic integration and generate convergence in cross-country income. The persistent income differences between European countries have been a cause for concern among policymakers and commentators in recent years. Generally, the response has been to push for further integration efforts. Our results show that not only has convergence in the EU stalled over the last three decades, but also raise the question of whether the space for effective further policy may be limited. The US, widely considered to be more economically integrated than Europe, has a similar level of income convergence. Furthermore, the trajectories for income dispersion have been similar over the course of the 20th century. Perhaps there are forces other than the types of policies pursued in Europe that drive convergence. Or, to the extent that integration policies can generate convergence, there are limits to what can be achieved.

In the following section, we consider a theory that is consistent with this evidence. The theory takes insights from economic geography and applies them to the study of income convergence across regions.

3. Theory

In this section, we present a stylized model of income differences across regions. It is designed to speak to differences across US states, or across European countries, for example. The objective is to provide a simple framework that nests a classical view

Eastern and Western Europe, and then their reintegration. Our objective is to look at the broad trend. In light of these differences, we find the common patterns across the US and Europe all the more striking.

of income convergence across regions and to supplement it with migration and an agglomeration force. The migration is between regions, for example between European countries. Migration changes the supply of labor and can affect wages in equilibrium. Agglomeration provides an incentive for economic activity to be concentrated, and thereby affects migration. We analyze how migration costs and agglomeration jointly influence the spatial distribution of labor and the resulting economic outcomes across regions.

3.1. Environment

There is a single period and two regions, denoted by $i = 1, 2$. Each region has an initial endowment of labor N_i^0 , an amount of an immutable factor of production X_i , and a given technology level Z_i . We normalize the total amount of labor to equal one, so $N_1^0 + N_2^0 = 1$. We include an immutable production factor to capture the idea that regions have different physical characteristics that cannot be changed and affect production. One way to think about this is that different regions have different amounts of usable land, or other geographic characteristics such as access to rivers or oceans that affect production.

There is a representative competitive firm in each region i . The firm hires L_i workers at wage rate w_i , K_i units of capital at rental rate r , and combines them with the immutable factor X_i to produce output with a Cobb-Douglas production function:

$$Y_i = \tilde{Z}_i L_i^\alpha K_i^\eta X_i^{1-\alpha-\eta},$$

with $\alpha, \eta \in (0, 1)$ and $\alpha + \eta < 1$. The productivity level of the firm is:

$$\tilde{Z}_i = Z_i N_i^\theta,$$

where N_i is the number of people in region i after migration has occurred, and $\theta \geq 0$.¹⁹ There are two components to productivity. The second term, N_i^θ , captures the agglomeration force. If $\theta > 0$ then the productivity of the region increases in its population, capturing the idea that there are benefits from having people located close to each other. If $\theta = 0$ then this term drops out and productivity is purely exogenous, as is the standard assumption. Z_i , which we will call *technology*, is exogenous and captures everything else that affects productivity. To be conservative about the strength of agglomeration effects, we assume that $\theta + \alpha + \eta < 1$ so that the production function has decreasing returns to scale in variable inputs even when the agglomeration force is accounted for.²⁰

There is a single representative household that allocates its members across the two regions. With N_i^0 denoting the initial mass of people in region i , let N_i denote the mass

19. As discussed in footnote 3, we are studying the case in which agglomeration always has non-negative effects on productivity.

20. Under reasonable values for other parameters, this rules out an equilibrium with everyone in a single region, which is less empirically relevant. This decreasing returns to scale provides the congestion force in the model.

of people allocated to region i after migration. The household receives labor income and the profits from firms in the two regions. It derives utility from total consumption, but incurs a disutility cost from reallocating workers away from their initial locations. We assume that the parameters of the model are such that any migration is towards region 1 ($N_1 \geq N_1^0$), without loss of generality.²¹ There is a per-person cost of moving between regions, $\tilde{\gamma} = \gamma(Y_2/N_2)$ with $\gamma \geq 0$. We assume that this cost scales with the per capita income of the country the migrants are leaving. Think of this as capturing the cost of leaving one's culture, family and friends, with the assumption that the value of these things scales with income.²² The household's preferences are:

$$U(C, N_1 - N_1^0) = C - \tilde{\gamma}(N_1 - N_1^0).$$

The second term on the right-hand side captures the minimum cost to the household of achieving a chosen allocation of workers across the regions.²³

In this environment, we think of the initial allocation of labor in each region as the state of the economy at a certain point in time, and the equilibrium after migration as the long-run outcome once there has been time for migration to occur. In some settings, one may want to have separate households in each region.²⁴ We abstract from this to keep the model simple and focus on the effects of the agglomeration force on the supply side, and the effects of the migration cost.

3.2. *Equilibrium*

The details of the solution are in the Appendix. Here we discuss the equilibrium of the model and the intuition for the solution.

Competitive equilibrium. Start by considering the case in which the populations of the regions are fixed, and migration is not allowed. If both regions optimise their choice of capital, and the resulting levels of income per capita are sufficiently different, then workers migrate to the higher income region (region one). We focus on parameter values for the model, such that this occurs.²⁵

In this case, migration increases the supply of labor in the higher income region, decreasing the marginal product and the wage. The increase in the population of the region increases the marginal product through the agglomeration force. However, due

21. This assumption is merely a labeling convention, with the region that will be the recipient of migrants given parameter values labeled as region 1.

22. If the cost does not scale with income, then as output increases, the relative size of the cost goes to zero and migration becomes costless.

23. Note that we are implicitly assuming that the household does not internalize the effect of its migration decision on the migration cost. We view this as the relevant case as the household is representing many individuals who make their decisions separately, and have no impact on aggregate output or migration.

24. We are abstracting, for example, from questions about whether the nationals of one country would want to limit the entrance of foreigners. We are doing this to focus on the economic effects of the forces in the model. Also, we are focusing on within-Europe migration, where there is no legal barrier to migration across countries.

25. The specific condition for this is provided in the Appendix.

to the assumption that $\theta + \alpha + \eta < 1$, this effect is more than offset by the decreases returns to scale of the production technology in capital and labor. This decreasing returns provides the congestion force that acts against all workers moving to one region, and incomes per capita for the two regions diverging. In the region of emigration, the supply of labor falls, and the wage increases. This process continues until the wage difference closes to the point at which it equals the migration cost:

$$w_1 = w_2 + \tilde{\gamma}.$$

Note that the cost of migration increases through this process. As the emigrant country becomes wealthier, it becomes more costly to leave.

In equilibrium, the following condition holds:

$$\frac{y_1}{y_2} = 1 + \frac{\gamma}{\alpha}, \quad (1)$$

where $y_i \equiv Y_i/N_i$ is per capita output. This says that the size of the difference in relative income depends on the parameter determining the migration cost, γ , and the returns to labor in the production function, α . If the migration cost is larger, it is more costly to move workers to the higher wage region, and the income gap between the two regions is larger in equilibrium. The gap decreases in the returns to labor because, with higher returns to labor, the marginal product falls less with each additional worker, so the average income gap closes more.

Agglomeration externality. In the competitive equilibrium the firms do not internalize that their labor demand decisions affect the total population in their region, and thereby change productivity. That is, they take $\tilde{Z}_i$ as given. To assess the impact of this, we solve a version of the model in which a planner makes the labor demand decision for firms and internalizes the agglomeration force.²⁶

The solution to this planner's problem has the same structure as the solution to the competitive equilibrium. The planner will move people to the higher income region up to the point at which a certain level of relative per capita income is attained:

$$\frac{y_1}{y_2} = 1 + \frac{\gamma}{\theta + \alpha}. \quad (2)$$

The planner's choices result in closer relative per capita incomes for the two regions than in the competitive equilibrium. This happens because the planner internalizes agglomeration gains from concentrating workers in a region. Specifically, the planner understands that the marginal product of labor diminishes more slowly than firms realize, so it chooses to migrate more workers to the high-income region and achieve closer per capita incomes for the two regions.

26. There is a second externality in the model, which is that the choices of agents determine Y_2/N_2 and this changes the migration cost. We focus on the effects of the agglomeration externality as this force is the subject of the analysis. This implies that we are not solving for the optimal allocation.

3.3. *Convergence with agglomeration and migration*

To think about convergence, we continue to focus on the case in which per capita income differs sufficiently under the initial endowment that there is migration in equilibrium.

Consider the standard theory of convergence within the model. The assumptions for this case are that, in the long run, technology is shared across countries so that $Z_1 = Z_2$, there is no agglomeration ($\theta = 0$), labor is immobile ($\gamma = \infty$), and there is no fixed factor of production ($X_i = 0$). In this setting, the standard result is that countries choose the same capital-to-labor ratio,²⁷ which is enough to achieve equal income per capita across the regions.

We establish three results from the model about how departures from these classical assumptions limit convergence.

RESULT 1. *If there is no migration and the regions have different fixed factors of production, or there is an agglomeration force and the populations of the regions differ, then shared technology between regions ($Z_1 = Z_2$) is not enough to equalize per capita income.*

The essence of the first observation is that geography matters for convergence. If two regions have different natural characteristics that affect their production capacities (captured in the model by the fixed production factor X_i), then if both regions have access to the same technology, their income per capita will not be equal. If one region has a bigger population than another and there is agglomeration ($\theta > 0$), then the same is true. The region with the bigger population will have a productivity advantage, and even if the regions have the same technology and the same fixed production factor, their incomes per capita will differ. These considerations provide two rationales for why we would see permanent income differences between regions in the long run, even when the other assumptions of the classical theory of convergence hold.

RESULT 2. *If there is costless migration ($\gamma = 0$), then per capita income is equalized across regions, regardless of any other differences between them, or the existence of agglomeration.*

The second observation tells us that migration is powerful for generating income convergence. In the model, no matter the nature or size of the differences between regions or the existence of agglomeration, if there is costless migration, then income per capita will be equalized across them. People will move to wherever wages are higher and this migration will cause the wage difference to decrease until eventually wages converge. Applying this to an example from the earlier empirical evidence, this says that if people were indifferent between living in Denmark or Spain, aside from real income differences, and there were no administrative barriers or financial costs of moving, then people would migrate from Spain to Denmark until the point at which the real income difference was eliminated.

27. Generally, we think about this happening through a gradual process of capital accumulation over time. In the present model, we are abstracting from these dynamics and focusing on the long-run case in which countries can attain their optimal capital-to-labor ratios.

RESULT 3. *If migration is costly ($\gamma > 0$), then per capita income is not equalized across regions. If there is also agglomeration, then the competitive equilibrium has less migration and larger income gaps between regions than the planner would choose.*

Once migration is costly, equation (1) implies that, regardless of the source of the initial income difference between regions, income is not equalized in equilibrium. It can be due to technology, population size, or the fixed production factor. There is still migration from the poorer to the richer region, however, this will stop before complete income convergence occurs. An income gap equal to the size of the migration cost will remain. While we do not distinguish between sources of the migration cost in the model, these will matter for whether incomplete convergence is efficient or not. To the extent that migration costs are due to personal tastes for where people like to live or actual financial costs of moving, the wage gap in the model would be efficient. To the extent that the cost is due to administrative barriers preventing people who want to move from doing so, the gap would be inefficiently large.²⁸

The second component of Result 3 is about the role of agglomeration. This generates an externality because, when firms choose their labor demand, they do not internalize that their demand for labor increases the population of their region, resulting in a productivity gain. The planner does account for this, and therefore places a higher marginal value on each worker moving to a region. This results in more migration and a smaller income gap between regions. This suggests that migration policy is a relevant consideration for income convergence.

RESULT 4. *The stronger the agglomeration force, the more migration there is.*

This result highlights the fact that agglomeration encourages the population to be more concentrated. If the agglomeration force is stronger, then the returns to concentrating people in one region are higher. So, more people need to move to the higher income region for the income difference to diminish to the point that it equals the migration cost. As equation (1) shows, the strength of agglomeration does not change the relative income of regions in equilibrium, but it does affect how much migration occurs.

As a final observation, we might often think about agglomeration encouraging migration to higher population regions. The model demonstrates that there are more considerations than this. Migration goes towards the region with higher output per capita. A higher population can contribute to this through agglomeration, but technology and the endowment of the immutable production factor are also relevant. So the model is flexible enough to account for why people might migrate to Switzerland (i.e. technology and fixed production factors), even though it has a small population.

28. We are abstracting from questions about the preferences of natives over the demographic composition of their population.

4. Migration and population distributions

To close the paper, we return to the data. A key feature of the model mechanism is that people migrate to higher income regions. While there is existing evidence of this (e.g. Mayda, 2010; Ortega and Peri, 2013), we verify that the data is consistent with this for Europe over the period that we are studying. We also compare the overall population concentrations within the EU and the US as an indicator of the extent of agglomeration.

4.1. Data

We add several datasets to those discussed in Section 2.1 for the analysis. In order to analyze migration patterns in Europe, we use data from the United Nations Department of Economic and Social Affairs on international migration stocks and country populations. This data provides information every five years for 1990–2024, and includes a breakdown by the origin and destination countries of migrants. We use this data to obtain the stock of people who have migrated from each European country in our sample to other European countries, as well as the breakdown of this number by destination country. For origin countries we use the same set of countries as in Section 2. For destinations we also include Ireland, Luxembourg, Norway and Switzerland because these are popular destinations. We focus on Europe for this part of the analysis since we do not have data on cross-state migration in the US covering our period of analysis.

In order to compare population density across the United States and Europe, we add population data at a more granular level. For Europe we use data at the NUTS 2 level from the Rosés and Wolf (2021) database. This provides harmonized data on population for each NUTS 2 region at approximately 10-year increments from 1900 to 2015.²⁹ We use data for 12 of the 14 European countries in our main sample, since the dataset does not cover Greece and Cyprus. This gives us 150 NUTS 2 regions across Europe.³⁰ To illustrate the level of disaggregation, Portugal has five NUTS 2 regions: North, Central, Lisbon, Alentejo and Algarve. The data for the US, from the BEA, is at the metropolitan statistical area (MSA) level. Specifically, we use information on the population for 2015. There are 384 MSAs. These do not include rural areas, but still cover the majority of the US population. For example, 86% of the US population lived in an MSA in 2015.

4.2. European country level migration

Our first step is to evaluate whether there is evidence that people from poorer countries in Europe are more likely to emigrate to other countries. From the perspective of the

29. The years covered by the data are 1900, 1910, 1925, 1938, 1950, 1960, 1970, 1980, 1990, 2000, 2010 and 2015.

30. For the UK, the dataset only provides information at the NUTS 1 level (more aggregated) so we use these regions for this country. Eight of our 150 sample regions are aggregated from two or three NUTS 2 regions to ensure consistent borders over time. To limit the influence of outliers, we exclude regions with fewer than 200 000 people. This removes Finnish region FI20 for all years, Italian region ITC2 for 1900–1960, and Netherlands region NL23 for 1900–1980.

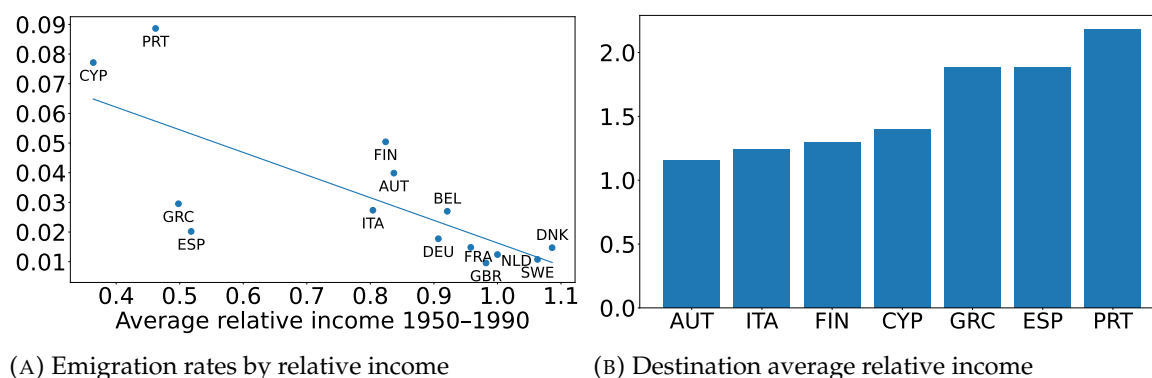


FIGURE 4: Emigration rates and destination country relative income, 1950–90

Notes: In panel (A), the horizontal axis is real GDP per capita, averaged over 1950–90, relative to the Netherlands. The vertical axis is the number of emigrants in other European countries in 1990, as a share of the origin country's average population for 1950–90. Ireland, Luxembourg, Norway and Switzerland are included as destination countries. Panel (B) shows the average income of destination countries relative to the origin country, weighted by the share of emigrants from the origin country that each destination country received.

model, if a country is poorer, then there is a greater economic incentive for its residents to move to wealthier countries. All else being equal, we would expect to see more people migrating. To evaluate this, we look at the European country-level data for 1950–90, because this was the period of our sample in which most of the convergence occurred (see Figure 3).³¹ Figure 4(A) considers the average income of each country in our sample relative to the Netherlands for 1950–90, and plots it against a measure of the total emigration rate to European destinations: the number of people from each country who are living abroad in Europe, as a share of the origin country's average population for 1950–90.

There is a strong negative correlation between relative income and the emigration rate, consistent with the theory. Cyprus and Portugal, in particular, are relatively poor and have high emigration rates of 8–9%. In the middle of the European income distribution, Italy, France and Austria have emigration rates of 3–5%, while the top income countries have emigration rates below 2%. There are exceptions to this pattern, with Greece and Spain being relatively poor but having much lower emigration rates than Cyprus and Portugal.

We are also interested in where emigrants move to. If migration contributes to convergence, then we should see that people move to higher income countries. Note that the model is not intended to be a full description of migration choices and, even within the model, migration does not have to be to the highest income region, because of migration costs. Thus, our aim is just to assess whether there is a general pattern of migration to higher income regions. For countries with high relative income, the scope to move to a higher income country is small, and even with modest migration costs the

31. We have also replicated the analysis for 1990–2015 for Eastern European countries that integrated with Western Europe after the fall of the Berlin wall. The results are consistent with those for our main sample.

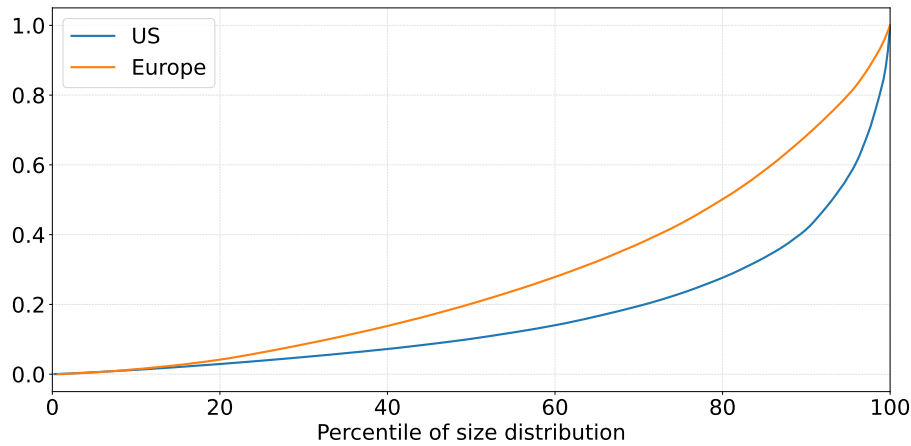


FIGURE 5: Cumulative population shares of US MSAs and European NUTS 2 regions, ranked by population size

Notes: The horizontal axis is that percentile of USA MSAs and European NUTS 2 regions in their (separate) population rankings. The vertical axis is the cumulative share of the population in the MSAs and NUTS 2 regions. All values are for 2015.

model would not predict such movements. We therefore focus on migration from the countries which had less than 90% of the income of the Netherlands on average from 1950–90. For each of these we compute the average relative income of the destination countries for migrants, weighting each destination by the share of migrants from the origin country that it received.³² The results are in Figure 4(B). We see that on average the emigrants from all of these countries moved to higher income destinations. Portugal stands out in this respect, with the average income of the destination countries being more than double the average income at home.³³

4.3. Population concentration

The second exercise compares the concentration of the populations in the US and Europe. To the extent that agglomeration effects depend on this, with the gains increasing as people are located closer together, we view the relative concentrations of the populations as an indicator of the extent of agglomeration in one area compared to the other.

This analysis uses US data at the MSA level and EU data at the NUTS 2 level.³⁴ To compare the spatial concentrations of the populations, we rank US MSAs and,

32. For example, if 60% of Portuguese emigrants went to country X, 40% went to country Y, and the relative incomes of these countries (compared to Portugal) averaged 1.8 and 2.2 over 1950–1990, respectively, then the average relative income of the destination countries would be $0.6 \times 1.8 + 0.4 \times 2.2 = 1.96$.

33. The primary destination of Portuguese emigrants in the period being studied was France, with 73% of emigrants going there.

34. These regional levels of disaggregation are not identical, making this exercise imperfect. The disaggregation for the US is finer, which we would expect to bias the US population distribution towards being more concentrated under our measure.

separately, European NUTS 2 regions by their populations for 2015. We plot the percentiles of these distributions against the corresponding cumulative population shares. The results for the US and Europe are in blue and orange, respectively, in Figure 5.

The US population is much more concentrated than Europe's, and the difference is substantial. In the US, more of the population is concentrated in the largest metropolitan areas. For example, the top 10% of metropolitan areas in the US account for 61% of the population, whereas in Europe it's only 31%. The top 5% in the US account for 43% of the population, while in Europe they account for 19%. Another way to quantify the difference is to compute how many people would need to move into Europe's 20 largest NUTS 2 regions for them to have the same population share as the top 20 MSAs in the US. Using 2015 data, 54.8 million people would need to move to the top 20 NUTS 2 regions to achieve this.

How should one interpret these differences in population concentration, in light of the similar levels of income per capita dispersion across US States and European countries in recent decades (recall Figure 3)? Through the lens of the model, the stable levels of income dispersion can be interpreted as the competitive equilibrium. Despite differences in population concentration, the US and Europe can have similar levels of income dispersion because they also differ in other dimensions. If Europe has smaller income differences across countries as a result of technology (Z) and immutable production factors (X), then it will achieve the same level of income dispersion with smaller differences in population across regions (i.e. a less concentrated population). One possible cause of this is that there are larger differences across US states in geographic characteristics such as weather and access to transportation routes that affect production. Another is that the US has more heterogeneity across states in research and development, and other innovative activities, than Europe has across countries.³⁵

5. Conclusion

This paper proposes an alternative perspective to the classic view on income convergence across countries. Instead of viewing convergence as the natural outcome of shared technology and the accumulation of production factors, we think about the extent of convergence as an equilibrium outcome when economic activity needs to be allocated over space. If there are agglomeration forces, geographic characteristics that affect production, and costly migration—all realistic features of the world, in our view—then we should expect income differences across countries. These arise even if technology is perfectly shared. Migration between countries helps to reduce income

35. Another potential dimension of heterogeneity between the US and Europe is migration costs. While these are difficult to measure, and it is beyond the scope of this paper to take on stand on their relative values, if migration costs were higher in Europe, then to observe a similar level of income dispersion in equilibrium, Europe would need to have a higher value of α . This implies a higher labor share. There is evidence for this in Gutiérrez and Piton (2020).

differences, but as long as migration is costly, income gaps should persist in the long run.

Data from US states and European countries is consistent with this view. Income per capita in both areas converged substantially over the last century, and, remarkably, the levels of dispersion have also been similar. Convergence has stalled since around 1980 in the US and 1990 in Europe, suggesting a stable equilibrium in both areas for over 30–40 years now. The theory rationalizes this as a result of income gaps no longer being large enough to induce migration, given the personal and financial costs of moving.

For policy makers interested in pursuing convergence, our theory suggests that a path consider is encouraging and supporting migration. The European population is substantially less concentrated than in the US, so there may be scope to attain additional gains from agglomeration if migration costs can fall. An important input into such considerations, would be additional research estimating the gains from agglomeration in Europe. There are, of course, also many other factors to consider that are outside the scope of this paper. Our hope is merely that our analysis provides useful insights for broader discussions of this topic.

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Appendix

Appendix A: Robustness exercises for Section 2

In this section we present three robustness exercises for the results in Section 2.

To address the concern that European countries and US states are not the same in terms of size, we replicate the analysis presented in Figure 2 at the level of US Census divisions. In 1980 the average population of the Census divisions was 25.1 million, compared to 25.2 million for our sample of European countries. Using the BEA dataset on the state-level personal income, we aggregate states into Census divisions following the official US Census Bureau classification. The regions are: New England, Middle Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain and Pacific. We then compute the real personal income per capita at the regional level relative to the Middle Atlantic. We choose this region as the reference point because it is relatively wealthy and includes New York, which was the benchmark state in Figure 2. The results are presented in Figure A.1 and closely mirror those in the main text.

A second potential concern is that the number of European countries is smaller than the number of US states, which could affect the estimate of the CV in Figure 3. While this is partially addressed by the standard deviation bands in the figure, we take this analysis a step further. To place the US on an equal footing with Europe, we perform a bootstrap exercise. We randomly drawing 14 states without replacement 500 times, compute the CV of personal income per state for each year for each sample, and then compute the mean and standard deviations of these estimates for each year. The results are presented in Figure A.2 and concord with the main analysis.

The third exercise recomputes the CVs for the US and Europe weighting each region by its population share each year instead of weighting regions equally.³⁶ The main effect is to decrease the CV for Europe, but the main messages from the analysis remain the same.

Appendix B: Model solution

This section provides some additional details for solving the model from Section 3.

Competitive equilibrium. In each region i , the representative firm chooses its labor input to maximize profits, taking the wage rate w_i , the rental rate r , productivity level $\tilde{Z}_i$ and the immutable production factor X_i as given. The choice variables are capital K_i labor L_i . The firm's profit maximization problem is:

$$\Pi_i = \max_{L_i, K_i} \left\{ \tilde{Z}_i L_i^\alpha K_i^\eta X_i^{1-\alpha-\eta} - w_i L_i - r K_i \right\}.$$

36. US states are weighted by their share of the US population each year, and European countries by their share of the total population of the European countries in the sample.

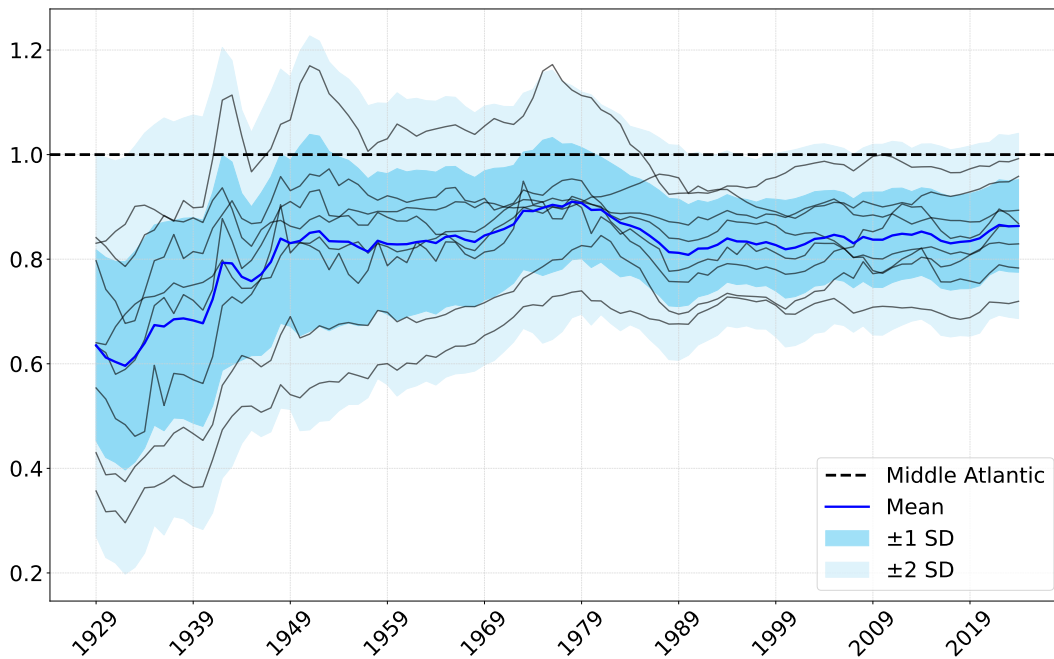


FIGURE A.1: US Census divisions personal income per capita relative to the Middle Atlantic.

Notes: Real person income per capita for each Census division is plotted with a black line. All values are relative to the Middle Atlantic for the relevant year. The solid blue line is the cross-regional mean in each year, and the shaded areas are ± 1 and ± 2 standard deviations, respectively.

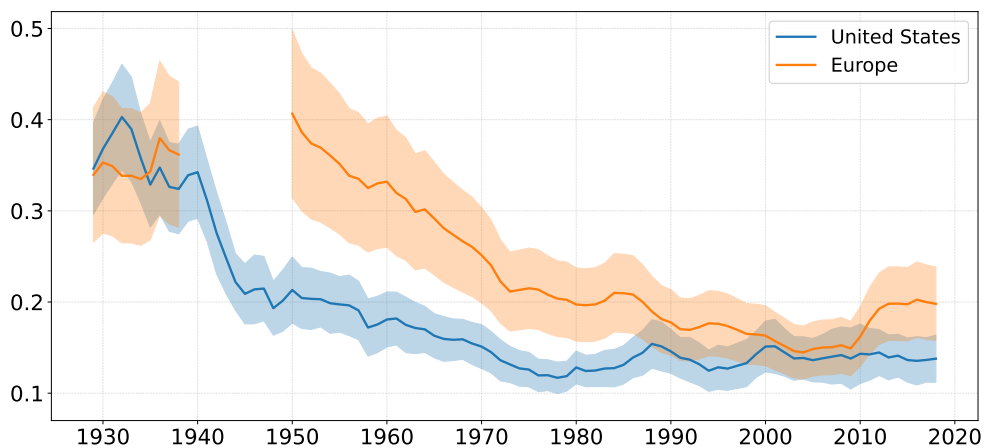


FIGURE A.2: Coefficient of variation for US state personal income per capita and European country GDP per capita—sample size robustness.

Notes: Personal income is real and GDP is real PPP. The coefficients of variation are computed year by year. For the US, the District of Columbia and Alaska are excluded. The European countries are Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Portugal, Spain, Sweden and the United Kingdom. The shaded areas are plus and minus one standard deviation confidence bands. For Europe they are computed following Feltz and Miller (1996). The US CV is estimated by randomly drawing 14 states without replacement 500 times and averaging the resulting CVs for each year. The blue shaded area represents the standard deviations of these estimates.

The representative household's problem is to allocate labor across regions to maximize utility, subject to the budget constraint and the total population of the two

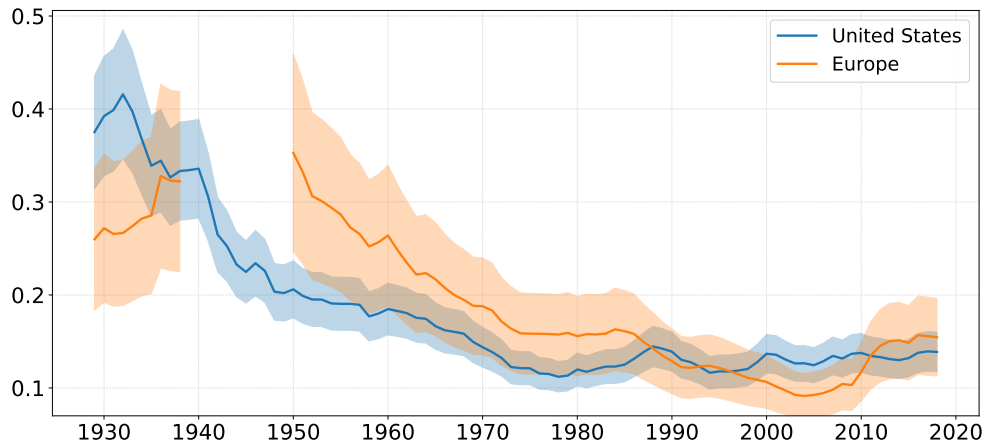


FIGURE A.3: Coefficient of variation for US state personal income per capita and European country GDP per capita—population weighted

Notes: Personal income is real and GDP is real PPP. The coefficients of variation are computed year by year, each observation weighted with the relative population within Europe or the United States. For the US, the District of Columbia and Alaska are excluded. The European countries are Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Italy, the Netherlands, Portugal, Spain, Sweden and the United Kingdom. The shaded areas are plus and minus one standard deviation confidence bands, computed following Feltz and Miller (1996).

regions:

$$\begin{aligned} \max_{N_1} U &= C - \tilde{\gamma}(N_1 - N_1^0) \\ \text{s.t. } C &= \Pi_1 + \Pi_2 + w_1 N_1 + w_2 N_2 + r(K_1 + K_2), \\ 1 &= N_1 + N_2. \end{aligned}$$

Initial conditions determine whether the competitive equilibrium features migration. If the two regions have endowments of Z_i , N_i^0 and X_i such that, given the optimal choice of capital, the marginal product of labor for the two regions is sufficiently close, then there will be no migration. In this case, output per capita will be equal across regions if the marginal product of labor given the optimal choice of capital is equal under the initial endowment, and otherwise it will differ. The condition determining whether there is migration in equilibrium is below.

If you take the first order condition of the household's problem, substitute in expressions for wages from the first order conditions of the firms' problems with respect to labor, and use the market clearing condition for labor ($N_1 + N_2 = 1$), then you can obtain

$$Z_1 N_1^{\theta+\alpha-1} K_1^\eta X_1^{1-\alpha-\eta} = Z_2 (1 - N_1)^{\theta+\alpha-1} K_2^\eta X_2^{1-\alpha-\eta} + \frac{\tilde{\gamma}}{\alpha}.$$

Our main equation for the competitive equilibrium, equation (1), follows from this.

The condition under which there will be no migration is

$$\frac{y_1^0}{y_2^0} \leq 1 + \frac{\gamma}{\alpha}, \quad (\text{B.1})$$

where y_i^0 is output per capita if a region produces with its initial endowment of labor and chooses capital optimally given this. This says that if the per capita incomes of the two regions are sufficiently close to begin with, then there is no migration.

Agglomeration externality. For the version of the economy in which a planner internalizes the agglomeration externality and chooses the allocation to maximize household utility, the problem:

$$\begin{aligned} \max_{N_1, K_1, K_2} \quad & U = Y_1 + Y_2 - \tilde{\gamma}(N_1 - N_1^0) \\ \text{s.t.} \quad & Y_i = \tilde{Z}_i L_i^\alpha K_i^\eta X_i^{1-\alpha-\eta}, \text{ for } i \in \{1, 2\}, \\ & \tilde{Z}_i = Z_i N_i^\theta, \text{ for } i \in \{1, 2\}, \\ & 1 = N_1 + N_2. \end{aligned}$$

To solve this problem, you can substitute the resource constraint for labor ($N_1 + N_2 = 1$) into the objective function of the planner to replace N_2 . Taking the first order condition with respect to labor yields

$$Z_1 N_1^{\theta+\alpha-1} K_1^\eta X_1^{1-\alpha-\eta} = Z_2 (1 - N_1)^{\theta+\alpha-1} K_2^\eta X_2^{1-\alpha-\eta} + \frac{\tilde{\gamma}}{\theta + \alpha},$$

and equation (2) follows from this. This is the solution for the case in which there is migration given the initial conditions. The planner, however, chooses no migration if

$$\frac{y_1^0}{y_2^0} \leq 1 + \frac{\gamma}{\theta + \alpha}.$$